

RYAN THOMAS

New Westminster, BC

 rta74@sfu.ca

 (778) - 986 - 1276

 <https://rta74.github.io/Ryan-Personal-Portfolio/>

EDUCATION

BASc in Mechatronics Systems Engineering
Simon Fraser University, Sept 2022 - Present

SKILLS

CAD

Assembly & modeling

- **Solidworks & Creo:** applied on multiple projects involving **GD&T**, **FEA analysis**, **sheet metal design**
- AutoCAD: drafted 2D blueprints and drawings using **FAI dimensions**

MECHANICAL

Analysis & Prototyping

- Implemented **DMFEA**, **root cause analysis** and **safety factor** on designs to address and fix issues
- Applied **statistical tolerance analysis** to 3D models, designs and assemblies
- Created and prototyped 3D models using **DFM**, **DFA** principles and **3D printing**

ELECTRICAL

Hands on experience with:

- **circuit design** and **analysis** with focus on circuit elements
- **SPA**, **oscilloscopes**, function generator, DMM, and power supply
- **Opamps**, **Diodes**, **Mosfets**, **BJT's**, and **energy storage devices**
- Soldering, Wiring

SOFTWARE

- **Ignition**, **C++**, **Matlab**, Altium, LTspice

TECHNICAL WORK EXPERIENCE

Maintenance Coordinator | Co-Op | Dometic

Sept 2025 - April 2026

- Led SCADA PM digitization by translating OEM manuals and practices into standardized maintenance tasks across 200+ machines & benches
- Assisted the Ignition SCADA scripting and logic while processing large maintenance datasets for automated tracking, distribution, and analysis
- Designed a custom air-oil filtration and ventilation solution to prevent oil condensation ingress in warehouse air-barrier intake systems
- Executed custom HVAC mechanical design in Creo using real-world measurements to resolve constrained installations and non-standard fit-ups

TECHNICAL PROJECTS

Hydroelectric Dam Design & Analysis | Fluid Mechanics | SFU

Jan - Apr 2025

- Performed full-cycle design analysis on a small-scale hydro dam, delivering modeling, evaluation, and recommendations to assess power generation potential and improve sustainability
- Applied Bernoulli's principle, Darcy-Weisbach equation and hydrostatic theory to calculate pressures, flow velocity, and head loss to model energy conversion and optimize system efficiency
- Estimated theoretical and actual power outputs based on fluid energy conversion and turbine efficiency; identified turbulence effects and recommended design improvements to reduce friction loss and enhance energy capture

Characterization of Active Components | Electrical | SFU

Sept 2024 - Dec 2024

- Conducted comprehensive active component analysis, characterizing diodes, BJT's, and MOSFETs to extract and validate their parameters
- Studied I-V characteristics of forward and reverse zener diode, integrated diodes within rectifier circuits with capacitors to validate behaviour
- Designed and implemented common emitter and collector BJT amplifiers, analyzed transistor behaviour in the active region, computing small signal gain and using LT Spice to run a frequency response simulation
- Used common source NMOS amplifier with capacitor coupling to obtain and compute small signal, Q point, frequency response, and voltage gain measurements

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TECHNICAL PROJECTS

Bike Frame Analysis | Mechanical, Structural Engineering | SFU

Oct - Dec 2023

- Simulated FEA through solidworks to assess frame fatigue strength under heavy loads and stress, identifying failure conditions and their feasibility
- Implemented design modifications to increase beam dimensions, resulting in reduced frame stress and a 2.5x increase to safety factor
- Calculated truss-beam stresses, forces, and safety factors for two theoretical frame conditions, using results to identify the safest design
- Conducted Root Cause Analysis on both frames to evaluate faults and potential failures, applying RCA principles to optimize improved design

Auto Fetch | Mechatronics, Product Design Engineering | SFU

Jan - Apr 2023

- Led the product design of an automatic fetch machine to help persons with disabilities engage with their dogs
- Utilized SolidWorks, applying GD&T for design validation and prototyping
- Applied Lean Manufacturing Principles to optimize design and production through strict time management and prioritizing key fixtures
- Improved DFM and DFA on assemblies by designing parts with minimal reorientation and self fastening features
- Conducted circuit design and analysis to ensure correct power usage, electrical safety and safe velocity and projectile control

WORK EXPERIENCE

Boston Pizza | Host Services, Part Time | New Westminster

June 2021 - Present

- Attend to phone and online orders-packing & confirming orders
- Support servers and management
- Assist clients and other miscellaneous guest services

EXTRACURRICULARS

Ice Hockey | New West Minor Hockey Association | New Westminster

Sept 2018 - April 2023

- Developed strong communication, teamwork, and interpersonal skills through consistent collaboration with teammates
- Took on a leadership role in final year mentoring players on and off the ice